Logic and Prolog – 2

Task 1

1. Create a Prolog file with the following facts

```
male(jack).
male(oliver).
male(ali).
male(james).
male(simon).
male(harry).
female(helen).
female(sophie).
female(jess).
female(lily).
parent_of(jack,jess).
parent_of(jack,lily).
parent_of(helen, jess).
parent_of(helen, lily).
parent_of(oliver,james).
parent_of(sophie, james).
parent_of(jess,simon).
parent_of(ali,simon).
parent_of(lily,harry).
parent_of(james,harry).
```

2. Complete the rules based on the above facts:

   father_of(X,Y):-

   mother_of(X,Y):-

   grandfather_of(X,Y):-

   grandmother_of(X,Y):-

   sister_of(X,Y):-

   aunt_of(X,Y):-

   brother_of(X,Y):-

   uncle_of(X,Y):-

   ancestor_of(X,Y):-

3. Write the appropriate queries and find the answers.

   a. Is Jack the father of Jess?
   b. Is Helen the mother of Lily?
   c. Are Jess and Lily sisters?
   d. Is Oliver the grandfather of Harry?
   e. Is Ali an ancestor of Harry?
   f. Who is the father of Simon?
   g. Who is the mother of James?
   h. Who are the children of Lily?

i. Who is the grandmother of Harry?
j. Who is the uncle of Harry?
k. Who is the aunt of Simon?
l. Does Harry have any brothers?
m. Find all father-child pairs in the tree:
n. Find all sisters in the family:
o. Who are all the descendants of Helen? (All people she is an ancestor to)
p. Who are all the ancestors of Simon?
q. Find all grandfathers and their respective grandchildren:
r. Find all males who are also fathers:
s. Who has both Jack and Helen as parents? (Finds full siblings)
t. Find anyone who is a parent but does NOT have any grandchildren in this tree:

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Task2

1. Create a Prolog file called airport_flights.pl and add the following facts to it

```
airport(london, uk).
airport(paris, france).
airport(frankfurt, germany).
airport(dubai, uae).
airport(delhi, india).
airport(singapore, singapore).
flight(ba101, london, paris, 1).
flight(lh202, paris, frankfurt, 1.5).
flight(ek303, frankfurt, dubai, 6).
flight(ek404, london, dubai, 7).
flight(ai505, dubai, delhi, 3.5).
flight(sq606, delhi, singapore, 5.5).
```

2. Add the following rules to the file:

```
direct_route(From, To) :-
        flight(_, From, To, _).

one_stop_route(From, To, Layover) :-
        flight(_, From, Layover, _),
        flight(_, Layover, To, _).

one_stop_duration(From, To, TotalTime) :-
        flight(_, From, Layover, Time1),
        flight(_, Layover, To, Time2),
        TotalTime is Time1 + Time2.

has_route(From, To) :-
        direct_route(From, To).
has_route(From, To) :-
        direct_route(From, Via),
        has_route(Via, To).
```

3. Answer the following questions with appropriate Prolog queries and find the answers.

    1. Is there a direct flight from London to Paris?
    2. Can I fly directly from Paris to Delhi?
    3. Is it possible to travel from London to Singapore eventually (via connections)?

4. Where can I fly to directly from London?
5. Which cities have a direct flight into Dubai?
6. What is the flight number of the flight leaving Delhi?
7. If I want to go from London to Dubai with exactly one stop, where will I layover?
8. Where can I travel to if I start in Frankfurt and make exactly one stop?
9. Find all direct flight routes and their durations:
10. How long will I spend in the air if I fly London -> Paris -> Frankfurt?
11. Find a 1-stop route from London to Delhi and calculate the total flying time:
12. List all pairs of cities that are connected by a 1-stop flight:
13. Find all direct flights that take longer than 5 hours:
14. Find short direct flights (under 2 hours):
15. Are there any flights where the destination country is the UK?
16. Find a flight that departs from a city in France:
17. Find all destinations reachable from London (direct or indirect):
18. Find any flight whose flight number starts with 'ek' (using wildcard sub-string check):
19. Find a destination you can reach from Paris that requires a total flying time of exactly 7.5 hours (with 1 stop):
20. Is there any city you can fly to from London that does NOT have a flight to Singapore?